

Quantitative OCT Angiography Evaluation of Peripapillary Retinal Circulation after Plaque Brachytherapy

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Purpose: To study peripapillary retinal capillary circulation in eyes treated with iodine 125 (¹²⁵I) plaque brachytherapy for uveal melanoma using optical coherence tomography angiography (OCTA).

Design: Cross-sectional study of 10 participants imaged with OCTA before uveal melanoma treatment and 15 participants imaged after development of radiation retinopathy, optic neuropathy, or both.

Participants: After institutional review board approval, participants were enrolled from an academic ocular oncology clinical practice. All participants had uveal melanoma in 1 eye, and treatment with ¹²⁵I plaque brachytherapy was planned or had taken place previously. Patients with low vision at baseline and uncontrolled hypertension were excluded. In the posttreatment group, 7 participants were men and 8 were women; age range was 38 to 81 years. Visual acuities in the irradiated eyes ranged from 20/20 to counting fingers. Visual acuities in the untreated fellow eyes were 20/25 or better.

Methods: Peripapillary retinal capillary circulation was measured by OCTA (Optovue, Inc). Optic disc scans measuring 4.5 × 4.5 mm were obtained.

Main Outcome Measures: The relationship of the peripapillary retinal capillary density (PPCD) as measured by OCTA to the calculated dose to the optic nerve (the dose to 50% of the disc [D50]) and visual acuity in logarithm of the minimal angle of resolution units were evaluated.

Results: No significant differences were observed in the PPCD as measured by OCTA when comparing the eye with melanoma with the fellow eye before brachytherapy; however, the PPCD was significantly lower in treated eyes (52.9% ± 22.4%) than in fellow eyes that did not receive radiation (73.3% ± 13.7%; $P = 0.004$). There was an inverse linear correlation between D50 and the PPCD (Pearson's $r = -0.528$; $P = 0.043$) and between visual acuity and the PPCD (Pearson's $r = -0.564$; $P = 0.028$).

Conclusions: Among patients with clinically apparent radiation retinopathy, radiation optic neuropathy, or both, PPCD was lower in the treated eye and correlated with the radiation dose to the optic nerve and the visual acuity. Optical coherence tomography angiography provides a measure of capillary changes after radiation and may serve as a quantitative end point to address visual prognosis. *Ophthalmology Retina* 2018;2:244-250 © 2017 by the American Academy of Ophthalmology



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Uveal melanomas are the most common primary intraocular malignancy in adults, with an incidence of approximately 5 to 6 cases per 1 million population.^{1–3} Eye-sparing treatment with radiation is an option for many patients diagnosed with uveal melanoma, but often leads to significant loss of vision. The degree of vision loss varies widely, with some patients experiencing only mild decline in measured visual acuity and others becoming entirely blind in the treated eye.^{4–7} In the landmark Collaborative Ocular Melanoma Study (COMS), 43% of patients treated with plaque brachytherapy were found to have measured visual acuity in the severely compromised range at 3 years after radiation.⁸

Vision loss after radiation therapy typically occurs because of radiation optic neuropathy and retinopathy caused

by vascular compromise leading to ischemia and edema.^{9–13}

Advanced vascular compromise can be observed with traditional fluorescein angiography, which shows retinal vascular leakage and capillary dropout in eyes with radiation-induced damage.^{14,15} Optical coherence tomography angiography (OCTA) is a new, noninvasive method for imaging retinal vasculature and providing quantitative information regarding ischemia in the retina.^{16–20} The association of radiation maculopathy with decreased parafoveal capillary density as measured by OCTA has been reported previously.²¹ Optical coherence tomography angiography findings may allow for earlier detection of radiation maculopathy.²² We hypothesized that OCTA can measure radiotherapy-induced changes in retinal vasculature associated with